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Assignment-2

Answer

Automated BI reporting (Dashboard & Scorecards)

Design approach

- Identify KPIs (Sales, Profit, growth)
- Choose tools (dashboard & Scorecard)
- Design simple and feractive UI
- Ensure real-time data Integration

Meta data Requirement

- Data Definitions (fields, tables)
- Data Source info
- Relationships between data
- Business rules

Automated Tasks

- Scheduled report generation
- Data refresh automatically
- Report distribution via email

Event based Alerts

- Trigger alerts when KPI crosses limit
- Example ↓ below target → alert.

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Answer 2

Features	Report	Dashboard	Scorecards
Purpose	Detailed data	Quick Overview	Performance tracki
Users	Analysis	managers	top Executives
Data	Static	Real-time	KPI-based
Example	Sales report	Sales dashboard	Balanced scorecard

Answer 3 Business Query & Reporting

Definition-

Process of Extracting data from decision making

Types of Queries

1) Simple Query → basic data retrieval
2) Complex Query → join, filters
3) Ad-hoc Query → user-defined
4) OLAP Query → multi dimensional analyses

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Role

- Helps managers analyze trends
- Supports strategic decisions
- Improves business performance

Answer 4

Types of BI Reports

1) Operational reports
   - Daily activities

   Examples → Attendance report

2) Tactical Report
   - Short term decisions

   Example – monthly sales

3) Strategic Reports
   - Long-term Planning

   Example – Annual growth report

4) Ad-hoc Reports
   – on demands reports

   Example – Common query report

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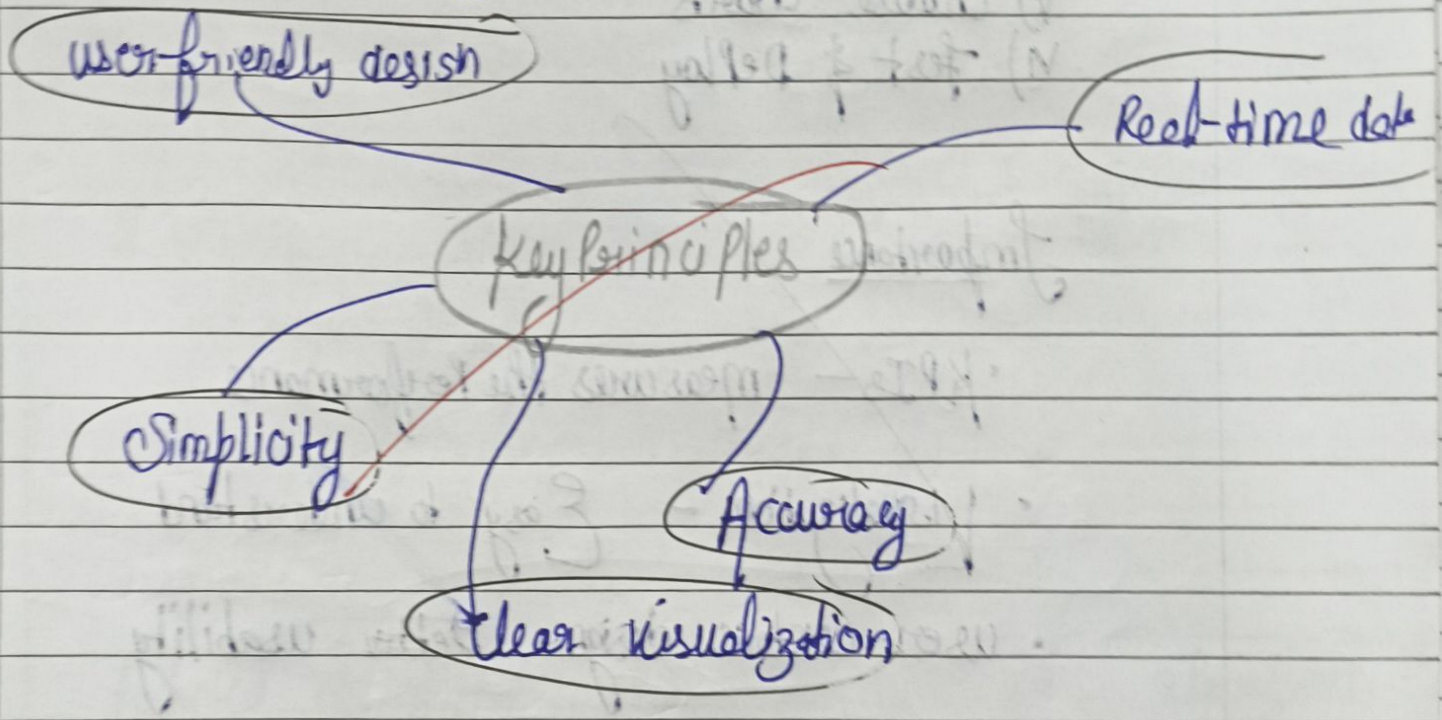
Answer 5

Dashboard & Scorecard concept

Dashboard is mainly used for visual display of data (charts, graphs)

Scorecard is reffers to tracks KPIs VS target



Conclusion- Dashboard shows the long-term result

Scorecard shows day to day Performance

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Answer 6

Steps in Dashboard Development

Step -

i) Define objectives
ii) Identify KPIs
iii) Collect data
iv) Design layout
v) Choose charts
vi) Test & Deploy

Importance

- KPIs - measures the Performance
- Visualization - Easy to understand
- user-Centric design - Better usability

Answer 7

Scorecard in BI

tool to measure performance using KPIs

Types of Scorecard

- Balanced Scorecard
- Operational Scorecard
- Strategic Scorecard.

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Improve decision making

Role

Compare actual vs target

Track Performance

Answer:

Metadata Model in BI

Definition - Data about dat called metadata

Types

Technical metadata (tables, schema)

Business metadata (definitions)

Operational metadata (logs, updates)

Importance

- Helps in query building
- Ensure data consistency
- Supports governance

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Answer:

Automated tasks & Event in BI

Automated tasks:

- Scheduled reports
- Data refresh
- Data backup

Events

Trigers actions based on Conditions

Benifits

- faster decision-making
- Reduce Error
- Saves time

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Answer 10

Role of metadata in Query & Reporting

Role:

- Defines data structure
- Help users understand data
- Improve query accuracy

Benifits

- Consistency
- Accuracy
- Easy data access
- Better usability.

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